

SENIOR AUTONOMY INTEGRATION ENGINEER - ROLE-ALIGNED RESUME

Pittsburgh, PA | 412.287.8640 | russelldudek@gmail.com | linkedin.com/in/russelldudek

Operator-technologist who makes complex systems legible and closes responsibility gaps across hardware, software, operations, quality, and stakeholders. Strongest when the visible symptom crosses formal ownership boundaries and the durable answer requires reproduction, evidence, coordinated repair, and a stronger system mechanism.

CORE CAPABILITIES

Cross-domain debugging | root cause and reproduction | robotics operations | configuration traceability | regression and quality systems | Python and TensorFlow | technical stakeholder alignment | Lean, Gemba, and standard work

MOST RELEVANT EXPERIENCE

VAPE-JET

2025-Present

Director of Operations

- Leads AI-first operating transformation across production, quality, support, engineering issue flow, ERP/Odoo, Jira, telemetry, and software-enabled robotics workflows.
- Connects factory, field, customer, and machine-fleet signals from approximately 1,000 fielded robots into prioritized work, earlier risk detection, cleaner handoffs, standard work, and accountability.

ZEUSVU

2018-2022

Chief Pilot, Autonomous Systems and Aerial Data Operations

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for aerial inventory and computer-vision field intelligence; connected sensing, mapping, GPS, risk, users, and operating evidence.

COMPUNETICS

2000-2013

Director of Operations and Engineering Management

- Led engineering, manufacturing, quality, and complex technical programs in advanced electronics and high-reliability environments; built root-cause practices, process controls, and quality systems.
- Collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD; published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

ADDITIONAL OPERATING EXPERIENCE

AMAZON

2014-2018

Operations Management / Site Leader

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
- Built throughput visibility, escalation paths, standard work, and operating routines across safety, quality, customer experience, and team engagement; contributed to Prime Air research and Amazon Flex geolocation concepts.

DUDEWORTH

2017-Present

Founder / AI and Automation Advisor

- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, and workflow design.

CARDINAL BUILDING PRODUCTS

2021-2023

Regional Director

- Built the company first remote regional operation and modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.

TECHNICAL TOOLKIT AND OPERATING METHODS

Python | TensorFlow | SQL | JavaScript | GIS and mapping | agentic AI | LLM workflows | RAG patterns | human-in-the-loop | evaluation and governance | ERP/Odoo | Jira | Confluence | HubSpot | Tableau

Root cause | failure reproduction | quality systems | process controls | standard work | Lean | Gemba | Six Sigma | escalation design | decision records | cross-functional alignment | training and adoption

EDUCATION AND TECHNICAL LEADERSHIP

University of Pittsburgh - Bachelor of Science, Chemistry; minors in Mathematics and Genetics. Former Chair, IEEE CMPT and EDS Pittsburgh Chapter. Industry publishing and technical speaking in advanced electronics and high-speed computing.

HONEST FIT STATEMENT

No verified senior production C++ tenure or prior automotive autonomy-stack title is claimed. The case is based on demonstrated cross-domain debugging, reliability, configuration-minded operations, robotics, autonomous systems, Python, and the ability to learn deeply through bounded technical work.